MTH251 & MTH252

Mathematical Analysis I & II

Course Textbook

Department of Mathematical Sciences

UNIST

Mathematical Analysis I & II

Course Textbook

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Source Code & Contributions
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Chapter 1

The Number Systems

1.1 The Natural, Integer, and Rational Number Systems

We denote the set $\{1, 2, 3, \ldots\}$ of natural numbers by $\mathbb{N}$. The set $\mathbb{N}$ is defined by the following axioms:

- $1 \in \mathbb{N}$.
- If $n \in \mathbb{N}$, then $s(n) \in \mathbb{N}$ where $s(n) = n + 1$ is called the **successor** of n.
- 1 is not the successor of any element of $\mathbb{N}$.
- If $m, n \in \mathbb{N}$ and $s(m) = s(n)$, then $m = n$.
- Any subset of $\mathbb{N}$ which contains 1, and contains $s(n)$ whenever it contains n, must be equal to $\mathbb{N}$.

The axioms above are called the **Peano axioms** (or *Peano Postulates*).

Remark. Let $P(n)$ be a statement about $n \in \mathbb{N}$, and let $S := \{n \in \mathbb{N} \mid P(n) \text{ is true}\}$. Assume that $P(n)$ satisfies the following conditions:

- $P(1)$ is true.
- $P(n + 1)$ is true whenever $P(n)$ is true.

Then, we have the following:

- $1 \in S$.
- $n + 1 \in S$ whenever $n \in S$.

Therefore, we have $S = \mathbb{N}$, i.e., $P(n)$ is true for all $n \in \mathbb{N}$. This method of proof is called the **principle of mathematical induction**.

Theorem 1.1.1. *Any nonempty subset of* $\mathbb{N}$ *has a least element. This theorem is called the* ***well-ordering property*** *of* $\mathbb{N}$.

Proof. Let S be a nonempty subset of $\mathbb{N}$. Assume that S does not have a least element. Let T and L be defined as follows:

- $T := \mathbb{N} \setminus S$.
- $L := \{n \in \mathbb{N} \mid \{1, 2, \ldots, n\} \subseteq T\}$.

We will show that $L = \mathbb{N}$, which implies that $S = \emptyset$, a contradiction.

- **Base case:** $1 \in L$ since $\{1\} \subseteq T$. Otherwise, $1 \in S$ and 1 is the least element of S.
- **Inductive step:** Assume that $n \in L \implies \{1, 2, \ldots, n\} \subseteq T$. Then $n + 1 \notin S$. Otherwise, $n + 1$ is the least element of S. Therefore, we have

  $$n + 1 \in T \implies \{1, 2, \ldots, n + 1\} \subseteq T.$$

  Thus $n + 1 \in L$.

By the principle of mathematical induction, we have $L = \mathbb{N}$. □

We define the set $\mathbb{Z}$ of integers and the set $\mathbb{Q}$ of rational numbers as follows:

- $\mathbb{Z} := \{m - n \mid m, n \in \mathbb{N}\}$.
- $\mathbb{Q} := \left\{\frac{m}{n} \,\middle|\, m \in \mathbb{Z},\ n \in \mathbb{N}\right\}$.

Also, we denote the set of positive numbers or negative numbers as follows:

- $\mathbb{Z}^+ := \{n \mid n \in \mathbb{N}\}$, $\mathbb{Z}^- := \{-n \mid n \in \mathbb{N}\}$.
- $\mathbb{Q}^+ := \left\{\frac{m}{n} \,\middle|\, m \in \mathbb{N},\ n \in \mathbb{N}\right\}$, $\mathbb{Q}^- := \left\{\frac{m}{n} \,\middle|\, m \in \mathbb{Z}^-,\ n \in \mathbb{N}\right\}$.

1.2 Ordered Sets and Ordered Fields

A set S is called an **ordered set** if it is equipped with a relation $<$ such that for any $x, y, z \in S$, we have the following properties:

(A) $x < y$ and $y < z$ implies $x < z$.

(B) Only one of the relations $x < y$, $x = y$, or $x > y$ holds.

$x > y$ means $y < x$, $x \leq y$ means ($x < y$ or $x = y$) and $x \geq y$ means $y \leq x$.

Remark. In an ordered set S, the relation $<$ is called an **order** (or a **total order**) on S. If a relation $<$ on T satisfies **(A)** only, then the relation $<$ is called a **partial order** on T and the set T is called **partially ordered**.

A set F is called a **field** if it is equipped with two operations, *addition* and *multiplication*, and satisfies the following axioms:

(A) Axioms for addition

- $x + y \in F$ for all $x, y \in F$.
- $x + y = y + x$ for all $x, y \in F$.
- $(x + y) + z = x + (y + z)$ for all $x, y, z \in F$.
- There exists an element $0 \in F$ such that $x + 0 = x$ for all $x \in F$.
- For each $x \in F$, there exists an element $-x \in F$ such that $x+(-x) = 0$.

(B) Axioms for multiplication

- $x \cdot y \in F$ for all $x, y \in F$.
- $x \cdot y = y \cdot x$ for all $x, y \in F$.
- $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ for all $x, y, z \in F$.
- There exists an element $1 \in F$ with $1 \neq 0$ such that $x \cdot 1 = x$ for all $x \in F$.
- For each $x \in F$ with $x \neq 0$, there exists an element $x^{-1} \in F$ such that $x \cdot x^{-1} = 1$.

(C) The distributive law

- $x \cdot (y + z) = x \cdot y + x \cdot z$ for all $x, y, z \in F$.

A set F is called an **ordered field** if it is both an ordered set and a field, and satisfies the following axioms:

- If $x, y, z \in F$ and $x < y$, then $x + z < y + z$.
- If $x, y \in F$ and $x > 0$ and $y > 0$, then $x \cdot y > 0$.

Definition 1.2.1. Let S be an ordered set and $E \subseteq S$.

- E is **bounded above** if there exists some $M \in S$ such that $x \leq M$ for all $x \in E$. Such an M is called an **upper bound** of E.
- E is **bounded below** if there exists some $m \in S$ such that $m \leq x$ for all $x \in E$. Such an m is called a **lower bound** of E.

Theorem 1.2.2. *Any nonempty subset of $\mathbb{Z}$ which is bounded below in $\mathbb{Z}$ has a unique least element, and any nonempty subset of $\mathbb{Z}$ which is bounded above in $\mathbb{Z}$ has a unique greatest element.*

Proof. Let E be a nonempty subset of $\mathbb{Z}$ which is bounded below in $\mathbb{Z}$. Then there exists a lower bound $m \in \mathbb{Z}$ of E. Let F be defined as follows:

$$F := \{x - m + 1 \mid x \in E\} \subseteq \mathbb{N}.$$

By Theorem 1.1.1, F has a least element n. Then we have

$$n \leq x - m + 1 \text{ for all } x \in E \implies n + m - 1 \leq x \text{ for all } x \in E.$$

Since $n \in F$, there exists a unique $x_0 \in E$ such that $n = x_0 - m + 1$, so $x_0 = n + m - 1$ is the unique least element of E.

Let S be a nonempty subset of $\mathbb{Z}$ which is bounded above in $\mathbb{Z}$. Then there exists an upper bound $M \in \mathbb{Z}$ of S. Let T be defined as follows:

$$T := \{-x + M + 1 \mid x \in S\} \subseteq \mathbb{N}.$$

By Theorem 1.1.1, T has a least element n. Then we have

$$n \leq -x + M + 1 \text{ for all } x \in S \implies M - n + 1 \geq x \text{ for all } x \in S.$$

Since $n \in T$, there exists a unique $x_0 \in S$ such that $n = -x_0 + M + 1$, so $x_0 = M - n + 1$ is the unique greatest element of S. □

Definition 1.2.3. Let S be an ordered set. Assume that $E \subseteq S$ is bounded above. If there exists an $\alpha \in S$ such that

- α is an upper bound of E.
- If $\beta < \alpha$, then β is not an upper bound of E.

then α is called the **supremum** (or *least upper bound*) of E, denoted by

$$\alpha = \sup E.$$

Similarly, Assume that $F \subseteq S$ is bounded below. If there exists an $\gamma \in F$ such that

- γ is a lower bound of F.
- If $\beta > \gamma$, then β is not a lower bound of F.

then γ is called the **infimum** (or *greatest lower bound*) of F, denoted by

$$\gamma = \inf F.$$

Example 1.2.4. Let $S \subseteq \mathbb{Q}^+$ be defined as

$$S \coloneqq \{p \in \mathbb{Q}^+ \mid p^2 < 2\}.$$

For any $p \in S$, let

$$q \coloneqq p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} > p.$$

Then we have $q \in \mathbb{Q}^+$ and

$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2} < 0 \implies q \in S.$$

This implies that any $p \in S$ is not an upper bound of S.

For any $p \in \mathbb{Q}^+$ with $p^2 > 2$, p is an upper bound of S. Let

$$q \coloneqq p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} < p.$$

Then we have $q \in \mathbb{Q}^+$ and

$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2} > 0 \implies q \text{ is a lower bound of } S.$$

This implies that such p is not the least upper bound of S. Since there exists no $p \in \mathbb{Q}^+$ such that $p^2 = 2$, S has no least upper bound in $\mathbb{Q}^+$.

Definition 1.2.5. An ordered set S is said to have the **least upper bound property** if any nonempty subset of S which is bounded above has a least upper bound in S.

Theorem 1.2.6. *Let S be an ordered set with the least upper bound property. Then, any nonempty subset of S which is bounded below has a greatest lower bound in S.*

Proof. Let E be a nonempty subset of S which is bounded below. Let L be defined as follows:

$$L \coloneqq \{\lambda \mid \lambda \text{ is a lower bound of } E\}.$$

Then, L is nonempty and bounded above. By the least upper bound property, there exists $\alpha = \sup L \in S$. We will show that $\alpha = \inf E$.

Assume that $\alpha \notin L$. Then we have the following:

- α is not a lower bound of $E \implies$ There exists $\beta \in E$ such that $\beta < \alpha$.

- $\alpha = \sup L \implies \beta$ is not an upper bound of $L \implies$ There exists $\gamma \in L$ such that $\beta < \gamma \leq \alpha$.
- $\gamma \in L \implies \gamma$ is a lower bound of $E \implies \gamma \leq \beta$.

This is a contradiction. Therefore, we have $\alpha \in L$, i.e., α is a lower bound of E. Since α is an upper bound of L, the following holds:

$$\text{For any } \delta > \alpha, \text{ we have } \delta \notin L \implies \delta \text{ is not a lower bound of } E.$$

Therefore, we can conclude that $\alpha = \inf E$. $\square$

1.3 The Real Field

There exists an ordered field $\mathbb{R}$ which has the least upper bound property. Moreover, $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield, i.e., $\mathbb{Q} \subseteq \mathbb{R}$ and the operations of addition and multiplication on $\mathbb{R}$ coincide with those on $\mathbb{Q}$.

We can construct $\mathbb{R}$ from $\mathbb{Q}$. But for now, we will assume the existence of $\mathbb{R}$ and its properties.

Theorem 1.3.1. *For any $x \in \mathbb{R}^+$ and $y \in \mathbb{R}$, there exists some $n \in \mathbb{N}$ such that $nx > y$. This theorem is called the **Archimedean property** of $\mathbb{R}$.*

Proof. Assume that $x, y \in \mathbb{R}$ and $x > 0$. Let E be defined as follows:

$$E := \{nx \mid nx \leq y,\ n \in \mathbb{N}\}.$$

If $E = \emptyset$, which implies that $x > y$, then $n = 1$ satisfies $nx > y$. Otherwise, E is nonempty and bounded above by y. By the least upper bound property, there exists $\alpha = \sup E \in \mathbb{R}$. Then we have the following:

- $\alpha - x$ is not an upper bound of $E \implies$ There exists some $m \in \mathbb{N}$ such that $mx > \alpha - x$.
- $mx > \alpha - x \implies (m+1)x > \alpha \implies (m+1)x \notin E \implies (m+1)x > y$.

Therefore, we can take $n = m + 1$ to complete the proof. $\square$

Corollary 1.3.2. *For any $\varepsilon \in \mathbb{R}^+$ and $b \in \mathbb{R}$ with $b > 1$, there exists some $n \in \mathbb{N}$ such that $b^{-n} < \varepsilon$.*

Proof. For any $n \in \mathbb{N}$ and $x \in \mathbb{R}$ with $x \geq -1$, we have

$$(1 + x)^n \geq 1 + nx.$$

Let $a \in \mathbb{R}^+$ and $\varepsilon \in \mathbb{R}^+$ be given. By Theorem 1.3.1, there exists some $n \in \mathbb{N}$ such that $na > \dfrac{1}{\varepsilon}$. Then we have

$$(1+a)^n \geq 1 + na > na > \frac{1}{\varepsilon} \implies (1+a)^{-n} < \varepsilon.$$

Since $b > 1$, we can take $a = b - 1 > 0$ to complete the proof. $\square$

Theorem 1.3.3. *Any nonempty subset of $\mathbb{Z}$ which is bounded below in $\mathbb{R}$ has a unique least element, and any nonempty subset of $\mathbb{Z}$ which is bounded above in $\mathbb{R}$ has a unique greatest element.*

Proof. Let E be a nonempty subset of $\mathbb{Z}$ which is bounded below in $\mathbb{R}$. Then there exists a lower bound $\alpha \in \mathbb{R}$ of E. Then we have the following:

- If $\alpha \geq 0$, then $-1 \in \mathbb{Z}$ is a lower bound of E.
- If $\alpha < 0$, then by Theorem 1.3.1, there exists some $n \in \mathbb{N}$ such that $n > -\alpha$, which implies that $-n \in \mathbb{Z}$ is a lower bound of E.

Therefore, E is bounded below in $\mathbb{Z}$. By Theorem 1.2.2, E has a unique least element.

Let F be a nonempty subset of $\mathbb{Z}$ which is bounded above in $\mathbb{R}$. Then there exists an upper bound $\beta \in \mathbb{R}$ of F. Then we have the following:

- By Theorem 1.3.1, there exists some $n \in \mathbb{N}$ such that $n > \beta$, which implies that $n \in \mathbb{Z}$ is an upper bound of F.

Therefore, F is bounded above in $\mathbb{Z}$. By Theorem 1.2.2, F has a unique greatest element. $\square$

Definition 1.3.4. For any $x \in \mathbb{R}$, the **floor function** $\lfloor x \rfloor$ and **ceiling function** $\lceil x \rceil$ are defined as follows:

$$\lfloor x \rfloor := \max\{k \in \mathbb{Z} \mid k \leq x\}, \quad \lceil x \rceil := \min\{k \in \mathbb{Z} \mid k \geq x\}$$

Theorem 1.3.3 guarantees the *existence* and *uniqueness* of $\lfloor x \rfloor$ and $\lceil x \rceil$.

Theorem 1.3.5. *For any $x, y \in \mathbb{R}$ with $x < y$, there exists some $p \in \mathbb{Q}$ such that $x < p < y$. This theorem states that $\mathbb{Q}$ is **dense** in $\mathbb{R}$.*

Proof. Let $\varepsilon := y - x > 0$. By Theorem 1.3.1, there exists some $n \in \mathbb{N}$ such that $\dfrac{1}{n} < \varepsilon$. Let S be defined as follows:

$$S := \left\{ k \in \mathbb{Z} \;\middle|\; \frac{k}{n} < x \right\}.$$

S is nonempty since $n\lfloor x \rfloor - 1 \in S$, and is bounded above by $nx \in \mathbb{R}$. By Theorem 1.3.3, S has a greatest element, say m.

$$x < \frac{m+1}{n} < x + \frac{1}{n} < x + (y - x) = y.$$

Therefore, we can take $p = \dfrac{m+1}{n} \in \mathbb{Q}$ to complete the proof. $\square$

Theorem 1.3.6. *For each $x \in \mathbb{R}^+$ and $n \in \mathbb{N}$, there exists a unique $y \in \mathbb{R}^+$ such that $y^n = x$. We denote y by $\sqrt[n]{x}$ or $x^{\frac{1}{n}}$.*

Proof. Let E be defined as follows:

$$E := \{t \in \mathbb{R}^+ \mid t^n < x\}.$$

E is nonempty since $2^{-1} \in E$ if $x \geq 1$, and $x^2 \in E$ otherwise. Also, E is bounded above by $\max\{x, 1\}$. By the least upper bound property, there exists $\alpha = \sup E \in \mathbb{R}^+$. We will show that $\alpha^n = x$.

For any $a, b \in \mathbb{R}^+$ with $a < b$, we have

$$b^n - a^n = (b - a)\sum_{k=0}^{n-1} b^{n-1-k}a^k < (b - a)nb^{n-1}.$$

Assume that $\alpha^n < x$. By Theorem 1.3.5, there exists some $h \in \mathbb{R}^+$ such that

$$h < \min\left\{1,\ \frac{x - \alpha^n}{n(\alpha + 1)^{n-1}}\right\}.$$

Put $a = \alpha$ and $b = \alpha + h$. Then we have

$$\begin{aligned} &(\alpha + h)^n - \alpha^n < hn(\alpha + h)^{n-1} < hn(\alpha + 1)^{n-1} \\ \Longrightarrow\ &(\alpha + h)^n < \alpha^n + hn(\alpha + 1)^{n-1} < \alpha^n + (x - \alpha^n) = x. \end{aligned}$$

Therefore, we have $\alpha + h \in E$, which is a contradiction.

Assume that $\alpha^n > x$. By Theorem 1.3.5, there exists some $h \in \mathbb{R}^+$ such that

$$h < \min\left\{1,\ \alpha,\ \frac{\alpha^n - x}{n(\alpha + 1)^{n-1}}\right\}.$$

Put $a = \alpha - h$ and $b = \alpha$. Then we have

$$\begin{aligned} &\alpha^n - (\alpha - h)^n < hn\alpha^{n-1} < hn(\alpha+1)^{n-1} \\ \implies &(\alpha - h)^n > \alpha^n - hn(\alpha+1)^{n-1} > \alpha^n - (\alpha^n - x) = x. \end{aligned}$$

Therefore, we have $\alpha - h$ is an upper bound of E, which is a contradiction.

Therefore, we have $\alpha^n = x$, which guarantees the *existence* of y. For any $\alpha, \beta \in \mathbb{R}^+$, we have $\alpha \neq \beta \implies \alpha^n \neq \beta^n$, which guarantees the *uniqueness* of y. □

Definition 1.3.7. Let $y \in [0,1)$ and $b \in \mathbb{N}$ be given with $b \geq 2$. A sequence $\{a_n\}_{n=1}^{\infty}$ is called the **greedy base b expansion** of y if for each $n \in \mathbb{N}$, we have

$$0 \leq y - \sum_{k=1}^{n} \frac{a_k}{b^k} < \frac{1}{b^n}$$

where $a_k \in S_b := \{0, 1, 2, \ldots, b-1\}$ for all $k \in \mathbb{N}$. Especially, if $b = 10$, the sequence is called the **greedy decimal expansion** of y.

Theorem 1.3.8. *For any $y \in [0,1)$, the greedy base b expansion of y exists and is unique.*

Proof. Let $\{t_n\}_{n=1}^{\infty}$, $\{T_n\}_{n=1}^{\infty}$ and $\{a_n\}_{n=1}^{\infty}$ be defined as follows:

- $t_1 := by$, $T_1 := \{t \in S_b \mid t \leq t_1\}$, $a_1 := \max T_1$.
- $t_2 := b^2 y - ba_1$, $T_2 := \{t \in S_b \mid t \leq t_2\}$, $a_2 := \max T_2$.

  $\vdots$
- $t_n := b^n y - \sum_{k=1}^{n-1} a_k b^{n-k}$, $T_n := \{t \in S_b \mid t \leq t_n\}$, $a_n := \max T_n$

  for all $n \in \mathbb{N}$ with $n \geq 2$,

Assume that $0 \leq t_n < b$ for some $n \in \mathbb{N}$. Then, since $0 \in T_n$ and T_n is bounded above, T_n has a greatest element by Theorem 1.2.2, which implies that $\{a_n\}_{n=1}^{\infty}$ is well-defined. Let $P(n)$ be the statement that

$$P(n) : 0 \leq t_n < b \text{ and } a_n \text{ is well-defined.}$$

- **Base case:** $P(1)$ is true since $0 \leq t_1 = by < b$.

- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$. Then we have

$$\begin{aligned} t_{n+1} &= b^{n+1}y - \sum_{k=1}^{n} a_k b^{n+1-k} \\ &= b\left(b^n y - \sum_{k=1}^{n} a_k b^{n-k}\right) \\ &= b\left(b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} - a_n\right) \\ &= b(t_n - a_n). \end{aligned}$$

  Since $a_n = \max T_n$ and $t_n < b$, we have

$$a_n \le t_n < a_n + 1 \implies 0 \le t_n - a_n < 1 \implies 0 \le t_{n+1} = b(t_n - a_n) < b.$$

  Thus $P(n+1)$ is true whenever $P(n)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$. By the definition of a_n, we have

$$\begin{aligned} & a_n \le b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} < a_n + 1 \\ \implies & 0 \le b^n y - \sum_{k=1}^{n} a_k b^{n-k} < 1 \\ \implies & 0 \le y - \sum_{k=1}^{n} \frac{a_k}{b^k} < \frac{1}{b^n} \end{aligned}$$

for each $n \in \mathbb{N}$. Therefore, the *existence* of the *greedy base b expansion* of y is guaranteed.

Next, assume that there exist two sequences $\{a_n\}_{n=1}^{\infty}$ and $\{c_n\}_{n=1}^{\infty}$ such that for each $n \in \mathbb{N}$, we have

- $0 \le y - \sum_{k=1}^{n} \frac{a_k}{b^k} < \frac{1}{b^n}$
- $0 \le y - \sum_{k=1}^{n} \frac{c_k}{b^k} < \frac{1}{b^n}$

where $a_k \in S_b$ and $c_k \in S_b$ for all $k = 1, 2, \ldots, n$. We will show that for each $n \in \mathbb{N}$, we have $a_n = c_n$. Let $Q(n)$ be the statement that

$$Q(n) : a_k = c_k \text{ for all } k = 1, 2, \ldots, n.$$

- **Base case:** We have
  - $0 \leq y - \dfrac{a_1}{b} < \dfrac{1}{b} \implies \dfrac{a_1}{b} \leq y < \dfrac{a_1+1}{b}$.
  - $0 \leq y - \dfrac{c_1}{b} < \dfrac{1}{b} \implies \dfrac{c_1}{b} \leq y < \dfrac{c_1+1}{b}$.

  By combining the two inequalities, we have
  - $\dfrac{a_1}{b} \leq y < \dfrac{c_1+1}{b} \implies a_1 < c_1 + 1 \implies a_1 \leq c_1$.
  - $\dfrac{c_1}{b} \leq y < \dfrac{a_1+1}{b} \implies c_1 < a_1 + 1 \implies c_1 \leq a_1$.

  Therefore, we have $a_1 = c_1$, which shows that $Q(1)$ is true.

- **Inductive step:** Assume that $Q(n)$ is true for some $n \in \mathbb{N}$. We have
  - $0 \leq y - \displaystyle\sum_{k=1}^{n+1} \frac{a_k}{b^k} < \frac{1}{b^{n+1}} \implies \sum_{k=1}^{n+1} \frac{a_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}}$.
  - $0 \leq y - \displaystyle\sum_{k=1}^{n+1} \frac{c_k}{b^k} < \frac{1}{b^{n+1}} \implies \sum_{k=1}^{n+1} \frac{c_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}}$.

  By combining the two inequalities, we have
  - $\displaystyle\sum_{k=1}^{n+1} \frac{a_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}} \implies \sum_{k=1}^{n+1} \frac{a_k - c_k}{b^k} < \frac{1}{b^{n+1}}$.
  - $\displaystyle\sum_{k=1}^{n+1} \frac{c_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}} \implies \sum_{k=1}^{n+1} \frac{c_k - a_k}{b^k} < \frac{1}{b^{n+1}}$.

  By the inductive hypothesis, we have $a_k = c_k$ for all $k = 1, 2, \ldots, n$. Therefore, we have
  - $\dfrac{a_{n+1} - c_{n+1}}{b^{n+1}} < \dfrac{1}{b^{n+1}} \implies a_{n+1} - c_{n+1} < 1 \implies a_{n+1} \leq c_{n+1}$.
  - $\dfrac{c_{n+1} - a_{n+1}}{b^{n+1}} < \dfrac{1}{b^{n+1}} \implies c_{n+1} - a_{n+1} < 1 \implies c_{n+1} \leq a_{n+1}$.

  Therefore, we have $a_{n+1} = c_{n+1}$, which shows that $Q(n+1)$ is true whenever $Q(n)$ is true.

By the principle of mathematical induction, $Q(n)$ is true for all $n \in \mathbb{N}$. Therefore, the *uniqueness* of the *greedy base* b *expansion* of y is guaranteed. □

Theorem 1.3.9. *Let $b \in \mathbb{N}$ be given with $b \geq 2$. Let $\{a_n\}_{n=1}^{\infty}$ be the greedy base b expansion of $a \in [0, 1)$, and let $\{c_n\}_{n=1}^{\infty}$ be the greedy base b expansion of $c \in [0, 1)$. If $a_n = c_n$ for all $n \in \mathbb{N}$, then we have $a = c$.*

Proof. Assume that $a \neq c$. Without loss of generality, Assume that $a < c$. Then, by Corollary 1.3.2, there exists some $n \in \mathbb{N}$ such that

$$b^n > \frac{1}{c-a} \implies \frac{1}{b^n} < c - a.$$

By the definition of the *greedy base b expansion*, we have

- $0 \leq a - \sum_{k=1}^{n} \frac{a_k}{b^k} < \frac{1}{b^n} \implies \sum_{k=1}^{n} \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n}.$
- $0 \leq c - \sum_{k=1}^{n} \frac{c_k}{b^k} < \frac{1}{b^n} \implies c < \sum_{k=1}^{n} \frac{c_k}{b^k} + \frac{1}{b^n}.$

By the assumption that $a_n = c_n$ for all $n \in \mathbb{N}$, we have

$$\sum_{k=1}^{n} \frac{a_k}{b^k} = \sum_{k=1}^{n} \frac{c_k}{b^k}.$$

Therefore, we have a contradiction:

$$\sum_{k=1}^{n} \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n} < c < \sum_{k=1}^{n} \frac{c_k}{b^k} + \frac{1}{b^n}.$$

Therefore, we have $a = c$, which completes the proof. □

Theorem 1.3.10. *Let $b \in \mathbb{N}$ be given with $b \geq 2$, and let $\{a_n\}_{n=1}^{\infty}$ be given with $a_k \in S_b$ for all $k \in \mathbb{N}$. If there exists no $y \in [0, 1)$ such that*

$$\{a_n\}_{n=1}^{\infty} \text{ is the greedy base } b \text{ expansion of } y,$$

then there exists some $N \in \mathbb{N}$ such that

$$a_n = b - 1 \quad \text{for all } n \geq N.$$

Proof. We will prove the contrapositive of the statement: For all $N \in \mathbb{N}$, if there exists some $n \in \mathbb{N}$ such that

$$n \geq N \text{ and } a_n \neq b - 1,$$

then there exists $y \in [0, 1)$ such that $\{a_n\}_{n=1}^{\infty}$ is the *greedy base b expansion* of y.

Assume that the condition of the above claim holds. Let $x_n := \sum_{k=1}^{n} \frac{a_k}{b^k}$ for each $n \in \mathbb{N}$. Since $a_k \in S_b$ for all $k \in \mathbb{N}$, we have

$$0 \leq x_n \leq \sum_{k=1}^{n} \frac{b-1}{b^k} = 1 - \frac{1}{b^n} < 1 \quad \text{for all } n \in \mathbb{N}.$$

Then, we have that $X := \{x_n \mid n \in \mathbb{N}\}$ is bounded above. By the least upper bound property, there exists $y = \sup X \geq 0$.

First, we will show that $y < 1$. By the assumption, there exists $m \in \mathbb{N}$ such that $a_m \neq b - 1$. Then, for all $n > m$, we have

$$\begin{aligned} x_n = x_m + \sum_{k=m+1}^{n} \frac{a_k}{b^k} &\leq x_m + \sum_{k=m+1}^{n} \frac{b-1}{b^k} \\ \implies x_n \leq x_m + \frac{1}{b^m} - \frac{1}{b^n} &< x_m + \frac{1}{b^m}. \end{aligned}$$

Since $a_m \leq b - 2$, we have

$$\begin{aligned} x_m + \frac{1}{b^m} &= \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{a_m + 1}{b^m} \\ &\leq \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{b-1}{b^m} \\ &\leq \sum_{k=1}^{m} \frac{b-1}{b^k} = 1 - \frac{1}{b^m}. \end{aligned}$$

We have $x_n \leq x_m$ for all $n \leq m$, and $x_n < x_m + \dfrac{1}{b^m} \leq 1 - \dfrac{1}{b^m}$ for all $n > m$. Thus $1 - \dfrac{1}{b^m}$ is an upper bound of X. Since $y = \sup X$, we have

$$y \leq 1 - \frac{1}{b^m} < 1 \implies y \in [0, 1).$$

Next, we will show that $\{a_n\}_{n=1}^{\infty}$ is the *greedy base b expansion* of y. Let $n \in \mathbb{N}$ be fixed. By the assumption, there exists $m \in \mathbb{N}$ such that $m \geq n$ and $a_m \leq b - 2$.

$$\begin{aligned} x_m = x_n + \sum_{k=n+1}^{m} \frac{a_k}{b^k} &< x_n + \sum_{k=n+1}^{m} \frac{b-1}{b^k} \\ \implies x_m < x_n + \frac{1}{b^n} - \frac{1}{b^m} &\implies x_m + \frac{1}{b^m} < x_n + \frac{1}{b^n}. \end{aligned}$$

Since $a_m \leq b - 2$, $x_m + \dfrac{1}{b^m}$ is an upper bound of X by the same argument as above. Since $y = \sup X$, we have

$$y \leq x_m + \frac{1}{b^m} < x_n + \frac{1}{b^n} \implies 0 \leq y - x_n < \frac{1}{b^n}.$$

Therefore, $\{a_n\}_{n=1}^{\infty}$ is the *greedy base b expansion* of y. $\square$

Theorem 1.3.11. *Let $b \in \mathbb{N}$ be given with $b \geq 2$ and let $\{a_n\}_{n=1}^{\infty}$ be given with $a_k \in S_b$ for all $k \in \mathbb{N}$. If there exists some $N \in \mathbb{N}$ such that*

$$a_n = b - 1 \quad \textit{for all } n \geq N,$$

then there exists no $y \in [0, 1)$ such that $\{a_n\}_{n=1}^{\infty}$ is the greedy base b expansion of y. This theorem is the converse of Theorem 1.3.10.

Proof. Assume that there exists some $y \in [0, 1)$ such that $\{a_n\}_{n=1}^{\infty}$ is the *greedy base b expansion* of y. Then we have

$$0 \leq y - \sum_{k=1}^{n} \frac{a_k}{b^k} < \frac{1}{b^n} \quad \text{for all } n \in \mathbb{N}.$$

Let $N \in \mathbb{N}$ be such that $a_n = b - 1$ for all $n \geq N$. Let $x_n := \sum_{k=1}^{n} \frac{a_k}{b^k}$ for each $n \in \mathbb{N}$. Then, for all $n \geq N$, we have

$$\begin{aligned} x_n &= x_{N-1} + \sum_{k=N}^{n} \frac{b-1}{b^k} = x_{N-1} + \frac{1}{b^{N-1}} - \frac{1}{b^n} \\ \implies y - x_n &= y - x_{N-1} - \frac{1}{b^{N-1}} + \frac{1}{b^n} \geq 0 \\ \implies \frac{1}{b^n} &\geq x_{N-1} + \frac{1}{b^{N-1}} - y. \end{aligned}$$

Since $0 \leq y - x_{N-1} < \frac{1}{b^{N-1}}$, we have $x_{N-1} + \frac{1}{b^{N-1}} - y > 0$. By Corollary 1.3.2, there exists some $n \in \mathbb{N}$ with $n \geq N$ such that

$$\frac{1}{b^n} < x_{N-1} + \frac{1}{b^{N-1}} - y,$$

which is a contradiction. Therefore, there exists no $y \in [0, 1)$ such that $\{a_n\}_{n=1}^{\infty}$ is the *greedy base b expansion* of y. $\square$

1.4 The Extended Real Number System

The Extended real number system $\overline{\mathbb{R}}$ is defined as

$$\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}.$$

We define the following operations on $\overline{\mathbb{R}}$:

- $x + (+\infty) := +\infty$ for all $x \in \mathbb{R} \cup \{+\infty\}$.
- $x + (-\infty) := -\infty$ for all $x \in \mathbb{R} \cup \{-\infty\}$.

- $x \cdot (\pm\infty) := \pm\infty$ for all $x \in \mathbb{R}^+ \cup \{+\infty\}$.
- $x \cdot (\pm\infty) := \mp\infty$ for all $x \in \mathbb{R}^- \cup \{-\infty\}$.
- $\dfrac{x}{\pm\infty} := 0$ for all $x \in \mathbb{R}$.

$\overline{\mathbb{R}}$ is not a field, but it is an ordered set with the following order relation:

$$-\infty < x < +\infty \quad \text{for all } x \in \mathbb{R}.$$

Theorem 1.4.1. *Any subset of $\overline{\mathbb{R}}$ has a least upper bound and a greatest lower bound in $\overline{\mathbb{R}}$.*

Proof. Let $E \subseteq \overline{\mathbb{R}}$. If $E = \emptyset$, then $\sup E = -\infty$ and $\inf E = +\infty$. If $E \neq \emptyset$, then we have the following cases:

- If E is bounded above in $\mathbb{R}$, then $\sup E$ exists in $\mathbb{R}$ by the least upper bound property of $\mathbb{R}$.
- If E is not bounded above in $\mathbb{R}$, then $\sup E = +\infty$.
- If E is bounded below in $\mathbb{R}$, then $\inf E$ exists in $\mathbb{R}$ by the greatest lower bound property of $\mathbb{R}$.
- If E is not bounded below in $\mathbb{R}$, then $\inf E = -\infty$.

$\square$

1.5 The Complex Field

The complex field $\mathbb{C}$ is defined as

$$\mathbb{C} := \{(x, y) \mid x, y \in \mathbb{R}\}.$$

We define the following operations on $\mathbb{C}$:

- $(x, y) + (u, v) := (x + u, y + v)$ for all $(x, y), (u, v) \in \mathbb{C}$.
- $(x, y) \cdot (u, v) := (xu - yv, xv + yu)$ for all $(x, y), (u, v) \in \mathbb{C}$.

$\mathbb{C}$ is a field with the following properties:

- The additive identity is $(0, 0)$ and the multiplicative identity is $(1, 0)$.
- The additive inverse of (x, y) is $(-x, -y)$ and the multiplicative inverse of (x, y) is
$$\left(\frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2}\right)$$
for all $(x, y) \in \mathbb{C}$ with $(x, y) \neq (0, 0)$.

Remark. The function

$$f : \{(x, 0) \mid x \in \mathbb{R}\} \subseteq \mathbb{C} \to \mathbb{R}$$

defined by $f((x, 0)) = x$ is an *isomorphism*. We can identify $\mathbb{R}$ with the subfield $\{(x, 0) \mid x \in \mathbb{R}\}$ of $\mathbb{C}$. We denote $(x, 0) \in \mathbb{C}$ by $x \in \mathbb{R}$.

Remark. We define $i = (0, 1) \in \mathbb{C}$. Then we have $i^2 = -1$, and any complex number $(x, y) \in \mathbb{C}$ can be written as

$$(x, y) = x + yi.$$

Definition 1.5.1. Let $z = x + yi \in \mathbb{C}$ for some $x, y \in \mathbb{R}$. The **real part** of z is defined as

$$\Re(z) = x \text{ or } \mathrm{Re}(z) = x,$$

the **imaginary part** of z is defined as

$$\Im(z) = y \text{ or } \mathrm{Im}(z) = y,$$

and the **complex conjugate** of z is defined as

$$\overline{z} = x - yi.$$

Theorem 1.5.2. *Let $z = x+yi \in \mathbb{C}$ and let $w = u+vi \in \mathbb{C}$ for some $x, y, u, v \in \mathbb{R}$. Then we have the following properties:*

- $\overline{(\overline{z})} = z$.
- $z = \overline{z} \iff z \in \mathbb{R}$.
- $\overline{z + w} = \overline{z} + \overline{w}$.
- $\overline{zw} = \overline{z} \cdot \overline{w}$.
- $z + \overline{z} = 2\mathrm{Re}(z)$ *and* $z - \overline{z} = 2i\mathrm{Im}(z)$.
- $z\overline{z} = x^2 + y^2 \in \mathbb{R}^+ \cup \{0\}$.

Proof. The proof follows directly from the definitions and properties of the operations on $\mathbb{C}$. $\square$

Definition 1.5.3. Let $z = x + yi \in \mathbb{C}$ for some $x, y \in \mathbb{R}$. the **modulus** of z is defined as

$$|z| = \sqrt{z\overline{z}} = \sqrt{x^2 + y^2} \in \mathbb{R}^+ \cup \{0\}.$$

The *existence* and *uniqueness* of $|z|$ follows from Theorems 1.3.6 and 1.5.2.

Theorem 1.5.4. *Let $a_1, a_2, \ldots, a_n \in \mathbb{C}$ and $b_1, b_2, \ldots, b_n \in \mathbb{C}$. Then we have*

$$\left|\sum_{k=1}^{n} a_k \overline{b_k}\right|^2 \leq \sum_{k=1}^{n} |a_k|^2 \cdot \sum_{k=1}^{n} |b_k|^2.$$

This theorem is called the ***Cauchy-Schwarz inequality****.*

Proof. Let $i, j \in \mathbb{N}$ with $1 \leq i < j \leq n$. Then we have

$$\begin{aligned} |a_i b_j - a_j b_i|^2 &= (a_i b_j - a_j b_i)\overline{(a_i b_j - a_j b_i)} \\ &= |a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2 - (a_i \overline{b_i} \cdot \overline{a_j} b_j + \overline{a_i} b_i \cdot a_j \overline{b_j}). \end{aligned}$$

Consider the following sums:

$$\begin{aligned} \sum_{i<j} (|a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2) &= \sum_{i<j} (|a_i|^2 |b_j|^2) + \sum_{j<i} (|a_i|^2 |b_j|^2) \\ &= \sum_{i \neq j} (|a_i|^2 |b_j|^2) \\ &= \sum_{k=1}^{n} |a_k|^2 \cdot \sum_{k=1}^{n} |b_k|^2 - \sum_{k=1}^{n} |a_k|^2 |b_k|^2. \end{aligned}$$

$$\begin{aligned} \sum_{i<j} (a_i \overline{b_i} \cdot \overline{a_j} b_j + \overline{a_i} b_i \cdot a_j \overline{b_j}) &= \sum_{i \neq j} (a_i \overline{b_i} \cdot \overline{a_j} b_j) \\ &= \sum_{i=1}^{n} a_i \overline{b_i} \cdot \sum_{j=1}^{n} \overline{a_j} b_j - \sum_{k=1}^{n} a_k \overline{b_k} \cdot \overline{a_k} b_k \\ &= \left|\sum_{k=1}^{n} a_k \overline{b_k}\right|^2 - \sum_{k=1}^{n} |a_k|^2 |b_k|^2. \end{aligned}$$

Therefore, we have

$$\sum_{i<j} |a_i b_j - a_j b_i|^2 = \sum_{k=1}^{n} |a_k|^2 \cdot \sum_{k=1}^{n} |b_k|^2 - \left|\sum_{k=1}^{n} a_k \overline{b_k}\right|^2 \geq 0.$$

□

Corollary 1.5.5. *Let $a_1, a_2, \ldots, a_n \in \mathbb{R}$ and $b_1, b_2, \ldots, b_n \in \mathbb{R}$. Then we have*

$$\left(\sum_{k=1}^{n} a_k b_k\right)^2 \leq \sum_{k=1}^{n} a_k^2 \cdot \sum_{k=1}^{n} b_k^2.$$

Theorem 1.5.6. *Let $z, w \in \mathbb{C}$. Then we have the following properties:*

(A) $|zw| = |z| \cdot |w|$.

(B) $|z + w| \leq |z| + |w|$.

(C) $|z| - |w| \leq |z - w|$.

Proof. Let $z = x + yi$ and $w = u + vi$ for some $x, y, u, v \in \mathbb{R}$.

(A) $|zw|^2 = (xu - yv)^2 + (xv + yu)^2 = (x^2 + y^2)(u^2 + v^2) = |z|^2 \cdot |w|^2$.

(B) By Corollary 1.5.5, we have $(xu + yv)^2 \leq (x^2 + y^2)(u^2 + v^2)$.

$$\begin{aligned} |z + w|^2 &= (x + u)^2 + (y + v)^2 \\ &= x^2 + y^2 + u^2 + v^2 + 2(xu + yv) \\ &\leq x^2 + y^2 + u^2 + v^2 + 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\ &= (|z| + |w|)^2. \end{aligned}$$

(C) Again, by Corollary 1.5.5, we have

$$\begin{aligned} |z - w|^2 &= (x - u)^2 + (y - v)^2 \\ &= x^2 + y^2 + u^2 + v^2 - 2(xu + yv) \\ &\geq x^2 + y^2 + u^2 + v^2 - 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\ &= (|z| - |w|)^2. \end{aligned}$$

□

1.6 Euclidean Spaces

For each $n \in \mathbb{N}$, we define the **Euclidean n-space** $\mathbb{R}^n$ as the set of all ordered n-tuples of real numbers, i.e,

$$\mathbb{R}^n := \{\boldsymbol{x} = (x_1, x_2, \ldots, x_n) \mid x_k \in \mathbb{R} \text{ for all } k = 1, 2, \ldots, n\}$$

where $x_1, x_2, \ldots, x_n$ are called the **coordinates** of $\boldsymbol{x}$. The elements of $\mathbb{R}^n$ with $n > 1$ are called **vectors** or *points*, which are denoted by boldface letters.

Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ for some $n \in \mathbb{N}$, and let $\alpha \in \mathbb{R}$. We define the following operations on $\mathbb{R}^n$:

- $\boldsymbol{x}+\boldsymbol{y} := (x_1+y_1, x_2+y_2, \ldots, x_n+y_n)$.
- $\alpha\boldsymbol{x} := (\alpha x_1, \alpha x_2, \ldots, \alpha x_n)$.

Then, $\mathbb{R}^n$ is a *vector space* over $\mathbb{R}$ with the zero vector $\mathbf{0} = (0, 0, \ldots, 0)$.

Definition 1.6.1. Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ for some $n \in \mathbb{N}$. The **inner product** of $\boldsymbol{x}$ and $\boldsymbol{y}$ is defined as

$$\boldsymbol{x} \cdot \boldsymbol{y} := \sum_{k=1}^{n} x_k y_k.$$

The **norm** of $\boldsymbol{x}$ is defined as

$$\|\boldsymbol{x}\| := \sqrt{\boldsymbol{x} \cdot \boldsymbol{x}} = \sqrt{\sum_{k=1}^{n} x_k^2}.$$

The **distance** between $\boldsymbol{x}$ and $\boldsymbol{y}$ is defined as

$$d(\boldsymbol{x}, \boldsymbol{y}) := \|\boldsymbol{x} - \boldsymbol{y}\| = \sqrt{\sum_{k=1}^{n} (x_k - y_k)^2}.$$

We can confirm that $d(x, y) = |x - y|$ in the case of $n = 1$.

Theorem 1.6.2. *Let $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}^n$ for some $n \in \mathbb{N}$. Then we have the following properties:*

(A) $|\boldsymbol{x} \cdot \boldsymbol{y}| \leq \|\boldsymbol{x}\| \cdot \|\boldsymbol{y}\|$.

(B) $\|\boldsymbol{x} + \boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$.

(C) $\|\boldsymbol{x} - \boldsymbol{y}\| \geq \big|\|\boldsymbol{x}\| - \|\boldsymbol{y}\|\big|$.

Proof. **(A)** By Corollary 1.5.5, we have

$$|\boldsymbol{x} \cdot \boldsymbol{y}|^2 = \left|\sum_{k=1}^{n} x_k y_k\right|^2 \leq \sum_{k=1}^{n} x_k^2 \cdot \sum_{k=1}^{n} y_k^2 = \|\boldsymbol{x}\|^2 \cdot \|\boldsymbol{y}\|^2.$$

(B) By part **(A)**, we have

$$\begin{aligned}\|\boldsymbol{x} + \boldsymbol{y}\|^2 = (\boldsymbol{x} + \boldsymbol{y}) \cdot (\boldsymbol{x} + \boldsymbol{y}) &= \|\boldsymbol{x}\|^2 + 2\boldsymbol{x} \cdot \boldsymbol{y} + \|\boldsymbol{y}\|^2 \\ &\leq \|\boldsymbol{x}\|^2 + 2|\boldsymbol{x} \cdot \boldsymbol{y}| + \|\boldsymbol{y}\|^2 \leq \|\boldsymbol{x}\|^2 + 2\|\boldsymbol{x}\| \cdot \|\boldsymbol{y}\| + \|\boldsymbol{y}\|^2 \\ &= (\|\boldsymbol{x}\| + \|\boldsymbol{y}\|)^2.\end{aligned}$$

(C) By part **(B)**, we have

$$\begin{aligned}\|\boldsymbol{x}\| = \|(\boldsymbol{x} - \boldsymbol{y}) + \boldsymbol{y}\| \leq \|\boldsymbol{x} - \boldsymbol{y}\| + \|\boldsymbol{y}\| &\implies \|\boldsymbol{x}\| - \|\boldsymbol{y}\| \leq \|\boldsymbol{x} - \boldsymbol{y}\|. \\ \|\boldsymbol{y}\| = \|(\boldsymbol{y} - \boldsymbol{x}) + \boldsymbol{x}\| \leq \|\boldsymbol{y} - \boldsymbol{x}\| + \|\boldsymbol{x}\| &\implies \|\boldsymbol{y}\| - \|\boldsymbol{x}\| \leq \|\boldsymbol{x} - \boldsymbol{y}\|.\end{aligned}$$

□

Chapter 2

Basic Topology

2.1 Relations

A **relation** $\sim$ on a set S defines a subset R of $S \times S$, and vice versa.

$$R \coloneqq \{(x, y) \in S \times S \mid x \sim y\}, \quad x \sim y \iff (x, y) \in R.$$

Definition 2.1.1. A relation $\sim$ on a set S is called an **equivalence relation** if it satisfies the following properties:

- Reflexivity: $x \sim x$ for all $x \in S$.
- Symmetry: If $x \sim y$, then $y \sim x$ for all $x, y \in S$.
- Transitivity: If $x \sim y$ and $y \sim z$, then $x \sim z$ for all $x, y, z \in S$.

Definition 2.1.2. A **partition** of a set S is a collection $\{A_\alpha\}_{\alpha \in I}$ of nonempty subsets of S such that

- $A_\alpha \cap A_\beta = \emptyset$ for all $\alpha, \beta \in I$ with $\alpha \neq \beta$.
- $\displaystyle\bigcup_{\alpha \in I} A_\alpha = S$.

Definition 2.1.3. Let $\sim$ be an equivalence relation on a set S. For each $x \in S$, the **equivalence class** of x is defined as

$$[x] \coloneqq \{y \in S \mid y \sim x\}.$$

We denote the set of all equivalence classes of S by $S/\sim \coloneqq \{[x] \mid x \in S\}$.

Theorem 2.1.4. *Let $\sim$ be an equivalence relation on a set S. For any $x, y \in S$, we have either*
$$[x] = [y] \quad \textit{or} \quad [x] \cap [y] = \emptyset.$$
Therefore, $S/\sim$ is a partition of S.

Proof. Assume that $[x] \cap [y] \neq \emptyset$. There exists some $z \in S$ such that $z \in [x]$ and $z \in [y]$. Then we have

$$z \sim x \text{ and } z \sim y \implies x \sim z \text{ and } z \sim y \implies x \sim y$$

by the symmetry and transitivity of $\sim$. Therefore, for any $w \in [x]$, we have

$$w \sim x \text{ and } x \sim y, \implies w \sim y \implies w \in [y].$$

This shows that $[x] \subseteq [y]$. Similarly, we can show that $[y] \subseteq [x]$. Therefore, we have $[x] = [y]$ whenever $[x] \cap [y] \neq \emptyset$. ☐

Theorem 2.1.5. *Let $\{A_\alpha\}_{\alpha \in I}$ be a partition of a set S. Let $\sim$ be a relation on S defined by*
$$x \sim y \iff x, y \in A_\alpha \textit{ for some } \alpha \in I.$$
Then, we have that $\sim$ is an equivalence relation on S, and the equivalence classes of $\sim$ are precisely the sets A_α.

Proof. We need to show that $\sim$ is reflexive, symmetric, and transitive.

- Reflexivity: For any $x \in S$, there exists $\alpha \in I$ such that $x \in A_\alpha$.
- Symmetry: If $x \sim y$, then there exists $\alpha \in I$ such that $y, x \in A_\alpha$.
- Transitivity: If $x \sim y$ and $y \sim z$, then there exist $\alpha, \beta \in I$ such that

  $$x, y \in A_\alpha \text{ and } y, z \in A_\beta.$$

  Since $y \in A_\alpha \cap A_\beta$ and the sets $\{A_\alpha\}_{\alpha \in I}$ are disjoint, we have $\alpha = \beta$, which implies that $x, z \in A_\alpha$.

Therefore, $\sim$ is an equivalence relation on S. The equivalence class of any $x \in S$ is precisely the set A_α containing x. ☐

2.2 Functions

A *function* (or a *mapping*) is defined with 3 components.

- A **domain** X.
- A **codomain** Y.

- A **rule** f which associates each element $x \in X$ to a *unique* element $y \in Y$.

f is said to be a *function* of X into Y, which is denoted by $f : X \to Y$. For each $x \in X$, the *unique* element $y \in Y$ associated with x is denoted by $f(x)$.

Remark. Let $f : X \to Y$ be a function. For any subset $E \subseteq X$, we define the **image** of E under f as

$$f(E) \coloneqq \{f(x) \mid x \in E\} \subseteq Y.$$

Especially, if $E = X$, then $f(X) \subseteq Y$ is called the **range** of f.

Remark. Let $f : X \to Y$ be a function. For any subset $E \subseteq Y$, we define the **inverse image** of E under f as

$$f^{-1}(E) \coloneqq \{x \in X \mid f(x) \in E\} \subseteq X.$$

Definition 2.2.1. Let $f : X \to Y$ be a function. We define the properties of f as follows:

- f is **injective** (or *one-to-one*) if for any $x_1, x_2 \in X$, we have
$$f(x_1) = f(x_2) \implies x_1 = x_2.$$
- f is **surjective** (or *onto*) if $f(X) = Y$.
- f is **bijective** (or a *one-to-one correspondence*) if it is both injective and surjective.

Definition 2.2.2. Let $f : X \to Y$ be a bijective function. Then, for any $y \in Y$, there exists a *unique* $x \in X$ such that $f(x) = y$. Therefore, we can define a function

$$f^{-1} : Y \to X, \quad f^{-1}(y) = x.$$

Such a function f^{-1} is called the **inverse function** of f.

Theorem 2.2.3. *Let $f : X \to Y$ be a bijective function. Then the inverse function $f^{-1} : Y \to X$ is bijective.*

Proof. Let $y_1, y_2 \in Y$ be given. Assume that $f^{-1}(y_1) = f^{-1}(y_2) = x$. Then, by Definition 2.2.2, we have

$$f(x) = y_1 \text{ and } f(x) = y_2 \implies y_1 = y_2$$

since f is a function. Therefore, we have that f^{-1} is injective. By the definition of image, we have that $f^{-1}(Y) = X$, which implies that f^{-1} is surjective. $\square$

Definition 2.2.4. Let $f : X \to Y$ be a function. Let $E \subseteq X$ and $F \subseteq Y$ be given with $f(E) \subseteq F$. We define the restriction functions as follows:

- The **restriction** of f to E is defined by
$$f|_E : E \to Y, \quad f|_E(x) = f(x) \quad \text{for all } x \in E.$$
- The **Corestriction** of f to F is defined by
$$f|^F : X \to F, \quad f|^F(x) = f(x) \quad \text{for all } x \in X.$$
- The **Bi-restriction** of f to E and F is defined by
$$f|_E^F : E \to F, \quad f|_E^F(x) = f(x) \quad \text{for all } x \in E.$$

We will simply call all of them as just *restriction*, if there is no ambiguity.

Theorem 2.2.5. *Let $f : X \to Y$ be a function. Let $E \subseteq X$ and $F \subseteq Y$ be given with $f(E) \subseteq F$. Then we have the following properties:*

(A) *If f is injective, then $f|_E^F$ is also injective.*

(B) *If $f(E) = F$, then $f|_E^F$ is surjective.*

(C) *If f is injective and $f(E) = F$, then $f|_E^F$ is bijective.*

Proof. **(C)** follows from **(A)** and **(B)**. Let $x_1, x_2 \in E$.

(A) Assume that f is injective. Then we have
$$f|_E^F(x_1) = f|_E^F(x_2) \implies f(x_1) = f(x_2) \implies x_1 = x_2.$$

(B) Assume that $f(E) = F$. Then, for any $y \in F$, there exists $x \in E$ such that $f(x) = f|_E^F(x) = y$. Therefore, we have $f|_E^F(E) = F$.

□

Definition 2.2.6. Let X, Y, Z be sets and $f : X \to Y$, $g : Y \to Z$ be functions. We define the **composition** of f and g as the function $g \circ f : X \to Z$ defined by
$$(g \circ f)(x) := g(f(x)) \quad \text{for all } x \in X.$$

Theorem 2.2.7. *Let X, Y, Z be sets and $f : X \to Y$, $g : Y \to Z$ be functions. Then we have the following properties:*

(A) *If f and g are injective, then $g \circ f$ is also injective.*

(B) *If f and g are surjective, then $g \circ f$ is also surjective.*

(C) *If f and g are bijective, then $g \circ f$ is also bijective.*

Proof. **(C)** follows from **(A)** and **(B)**. Let $x_1, x_2 \in X$ be given.

(A) Assume that f and g are injective. Then we have

$$\begin{aligned}(g \circ f)(x_1) = (g \circ f)(x_2) &\implies g(f(x_1)) = g(f(x_2)) \\ \implies f(x_1) = f(x_2) &\implies x_1 = x_2.\end{aligned}$$

(B) Assume that f and g are surjective. Then, for any $z \in Z$, there exists $y \in Y$ such that $g(y) = z$. Since f is surjective, there exists $x \in X$ such that $f(x) = y$. Therefore, we have $(g \circ f)(x) = g(f(x)) = g(y) = z$.

□

Theorem 2.2.8. *Let S be a set of sets. For Sections 2.3 and 2.4, we consider the relation $\sim$ defined on S by*

$$X \sim Y \iff \text{There exists a bijection } f : X \to Y.$$

The relation $\sim$ is an equivalence relation on S.

Proof. Let $X, Y, Z \in S$ be given.

- **Reflexivity:** The identity function $id_X : X \to X$, defined by $id_X(x) = x$ for all $x \in X$, is bijective. Therefore, we have $X \sim X$.
- **Symmetry:** Assume that $X \sim Y$. Then there exists a bijection $f : X \to Y$. By Theorem 2.2.3, the inverse function $f^{-1} : Y \to X$ is also bijective. Therefore, we have $Y \sim X$.
- **Transitivity:** Assume that $X \sim Y$ and $Y \sim Z$. Then there exist bijections $f : X \to Y$ and $g : Y \to Z$. By Theorem 2.2.7, the composition $g \circ f : X \to Z$ is also bijective. Therefore, we have $X \sim Z$.

□

2.3 Finite Sets

Let J_n be a set defined by

$$J_n := \{1, 2, \ldots, n\} \quad \text{for each } n \in \mathbb{N}.$$

A set X is said to be **finite** if $J_n \sim X$ for some $n \in \mathbb{N}$. In this case, n is called the **cardinality** of X, denoted by $|X|$. Especially, if $X = \emptyset$, we define $|X| = 0$. X is said to be **infinite** if X is not finite.

Lemma 2.3.1. *Any subset of J_n is finite for each $n \in \mathbb{N}$.*

Proof. Let $P(n)$ be the statement that any subset of J_n is finite.

- **Base case:** $J_1 = \{1\}$ has $\emptyset$ and J_1 as its only subsets. We have $J_1 \sim J_1$ by $f \equiv id_{J_1}$, and $\emptyset$ is finite by definition. Thus $P(1)$ is true.
- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$. Let $A \subseteq J_{n+1}$ be given. We have two cases:
  - If $n+1 \notin A$, then $A \subseteq J_n$, which is finite by the inductive hypothesis.
  - If $n+1 \in A$, then $A = B \cup \{n+1\}$ for some $B \subseteq J_n$. By the inductive hypothesis, B is finite, so there exists a bijection $f : J_m \to B$ for some $m \in \mathbb{N}$. Let $g : J_{m+1} \to A$ be defined by

    $$g(k) = \begin{cases} f(k) & \text{if } k \leq m \\ n+1 & \text{if } k = m+1 \end{cases}$$

    Then g is bijective, which implies that A is finite.

  Thus $P(n+1)$ is true whenever $P(n)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$. □

Theorem 2.3.2. *Any subset of a finite set is finite.*

Proof. Let X be a finite set and $Y \subseteq X$. Then there exists a bijection $f : J_n \to X$ for some $n \in \mathbb{N}$. Let Z be defined by

$$Z := f^{-1}(Y) \subseteq J_n.$$

Since $Z \subseteq J_n$, Z is finite by Lemma 2.3.1, so there exists a bijection $g : J_m \to Z$ for some $m \in \mathbb{N}$.

By Theorem 2.2.5, $f|_Z^Y$ is bijective. Then, by Theorem 2.2.7, the composition $f|_Z^Y \circ g : J_m \to Y$ is also bijective. Therefore, Y is finite. □

Lemma 2.3.3. *Let J_n be given for some $n \in \mathbb{N}$. If there exists an surjection $f : J_n \to Y$, then Y is finite.*

Proof. Let $f : J_n \to Y$ be a surjective function. For any $y \in Y$, let N_y be a set defined by

$$N_y := f^{-1}(\{y\}) \subseteq J_n.$$

Since f is surjective, N_y is nonempty. By Theorem 1.1.1, N_y has a least element, say n_y. Let M_Y be a set defined by

$$M_Y := \{n_y \mid y \in Y\} \subseteq J_n.$$

Let $g : M_Y \to Y$ be a function defined by

$$g(n_y) = y \quad \text{for each } n_y \in M_Y.$$

Then g is bijective. By Lemma 2.3.1, M_Y is finite, so there exists a bijection $h : J_m \to M_Y$ for some $m \in \mathbb{N}$. Then, by Theorem 2.2.7, the composition $g \circ h : J_m \to Y$ is bijective. Therefore, Y is finite. □

Theorem 2.3.4. *Let X be a finite set. If there exists an surjection $f : X \to Y$, then Y is finite.*

Proof. Let $f : X \to Y$ be a surjective function. Since X is finite, there exists a bijection $g : J_n \to X$ for some $n \in \mathbb{N}$. Then, by Theorem 2.2.7, the composition $f \circ g : J_n \to Y$ is surjective. Therefore, by Lemma 2.3.3, Y is finite. □

Theorem 2.3.5. *Let X be a finite ordered set. Then X has both a least element and a greatest element.*

Proof. Since X is finite, there exists a bijection $f : J_n \to X$ for some $n \in \mathbb{N}$. The elements of X can be listed as $X = \{a_1, a_2, \dots, a_n\}$, where $a_k = f(k)$ for each $k = 1, 2, \dots, n$.

First, we will show that X has a least element. Let $P(n)$ be the statement that any finite ordered set of cardinality n has a least element.

- **Base case:** $X = \{a_1\}$ has a least element a_1. Thus $P(1)$ is true.
- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$. Let X be a finite ordered set of cardinality $n + 1$ such that

  $$X = \{a_1, a_2, \dots, a_{n+1}\}.$$

  By the inductive hypothesis, a finite ordered subset $Y \subseteq X$ of cardinality n such that

  $$Y = \{a_1, a_2, \dots, a_n\}$$

  has a least element, say a_k for some $k = 1, 2, \dots, n$. We have two cases:

  - If $a_k \leq a_{n+1}$, then a_k is the least element of X.
  - If $a_{n+1} < a_k$, then a_{n+1} is the least element of X.

Thus $P(n+1)$ is true whenever $P(n)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$.

Next, we will show that X has a greatest element. Let $Q(n)$ be the statement that any finite ordered set of cardinality n has a greatest element.

- **Base case:** $X = \{a_1\}$ has a greatest element a_1. Thus $Q(1)$ is true.
- **Inductive step:** Assume that $Q(n)$ is true for some $n \in \mathbb{N}$. Let X be a finite ordered set of cardinality $n+1$ such that

$$X = \{a_1, a_2, \ldots, a_{n+1}\}.$$

By the inductive hypothesis, a finite ordered subset $Y \subseteq X$ of cardinality n such that

$$Y = \{a_1, a_2, \ldots, a_n\}$$

has a greatest element, say a_k for some $k = 1, 2, \ldots, n$. We have two cases:

  - If $a_k \geq a_{n+1}$, then a_k is the greatest element of X.
  - If $a_{n+1} > a_k$, then a_{n+1} is the greatest element of X.

Thus $Q(n+1)$ is true whenever $Q(n)$ is true.

By the principle of mathematical induction, $Q(n)$ is true for all $n \in \mathbb{N}$. □

Lemma 2.3.6. *Let J_m, J_n be given for some $m, n \in \mathbb{N}$. Then we have the following properties:*

(A) *If there exists an injection $f : J_m \to J_n$, then we have $m \leq n$.*

(B) *If there exists a surjection $f : J_m \to J_n$, then we have $m \geq n$.*

(C) *If there exists a bijection $f : J_m \to J_n$, then we have $m = n$.*

*This lemma guarantees the uniqueness of the cardinality. If $J_m \sim X$ and $J_n \sim X$, then we have $J_m \sim J_n$, which implies that $m = n$ by **(C)**.*

Proof. We will prove **(A)** by mathematical induction on m, **(B)** by mathematical induction on n, and **(C)** follows from **(A)** and **(B)**.

(A) Let $n \in \mathbb{N}$ be fixed. Let $P(m)$ be the statement that if there exists an injective function $f_m : J_m \to J_n$, then we have $m \leq n$.

- **Base case:** Let $m = 1$. Then $m \leq n$ holds. Thus $P(1)$ is true.

- **Inductive step:** Assume that $P(m)$ is true for some $m \in \mathbb{N}$. If $n = 1$, there does not exist an injective function $f_{m+1} : J_{m+1} \to J_n$ since $f_{m+1}(1) \neq f_{m+1}(m+1)$, where $P(m+1)$ is true.

  Otherwise, assume that such a function f_{m+1} exists. We have two cases:
  - If $f_{m+1}(m+1) = n$, let f_m be a function defined by
  $$f_m \equiv f_{m+1}|_{J_m}^{J_{n-1}} : J_m \to J_{n-1}.$$
  Then f_m is injective by Theorem 2.2.5. By the inductive hypothesis, we have $m \leq n-1$, which implies that $m+1 \leq n$.
  - If $f_{m+1}(m+1) = x \neq n$, let $g_n : J_n \to J_n$ be defined by
  $$g_n(y) = \begin{cases} x, & \text{if } y = n \\ n, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}$$
  Then g_n is bijective. Let f_m be a function defined by
  $$f_m \equiv (g_n \circ f_{m+1})|_{J_m}^{J_{n-1}} : J_m \to J_{n-1}.$$
  Then f_m is injective by Theorems 2.2.5 and 2.2.7. By the inductive hypothesis, we have $m \leq n-1$, which implies that $m+1 \leq n$.

  Therefore, $P(m+1)$ is true whenever $P(m)$ is true.

By the principle of mathematical induction, $P(m)$ is true for all $m \in \mathbb{N}$.

(B) Let $m \in \mathbb{N}$ be fixed. Let $Q(n)$ be the statement that if there exists a surjective function $f_n : J_m \to J_n$, then we have $m \geq n$.

- **Base case:** Let $n = 1$. Then $m \geq n$ holds. Thus $Q(1)$ is true.
- **Inductive step:** Assume that $Q(n)$ is true for some $n \in \mathbb{N}$. If $m = 1$, there does not exist a surjective function $f_{n+1} : J_m \to J_{n+1}$ since $f_{n+1}(1)$ is unique, where $Q(n+1)$ is true.

  Otherwise, assume that such a function f_{n+1} exists. We have two cases:
  - If $f_{n+1}(m) = n+1$, let K_{n+1} be a set defined by
  $$K_{n+1} \coloneqq (f_{n+1})^{-1}(\{n+1\}) \subseteq J_m.$$
  Let $f_n : J_m \to J_n$ be a function defined by
  $$f_n(y) = \begin{cases} f_{n+1}(y), & \text{if } y \in J_m \setminus K_{n+1} \\ n, & \text{if } y \in K_{n+1} \end{cases}$$
  Then f_n is surjective. By the inductive hypothesis, we have $m-1 \geq n$, which implies that $m \geq n+1$.

– If $f_{n+1}(m) = x \neq n+1$, let $g_{n+1} : J_{n+1} \to J_{n+1}$ be a function defined by

$$g_{n+1}(y) = \begin{cases} x, & \text{if } y = n+1 \\ n+1, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}$$

Then g_{n+1} is bijective. Let K_{n+1} be a set defined by

$$K_{n+1} \coloneqq (g_{n+1} \circ f_{n+1})^{-1}(\{n+1\}) \subseteq J_m.$$

Let $f_n : J_m \to J_n$ be a function defined by

$$f_n(y) = \begin{cases} (g_{n+1} \circ f_{n+1})(y), & \text{if } y \in J_m \setminus K_{n+1} \\ n, & \text{if } y \in K_{n+1} \end{cases}$$

Then f_n is surjective by Theorem 2.2.7. By the inductive hypothesis, we have $m-1 \geq n$, which implies that $m \geq n+1$.

Thus $Q(n+1)$ is true whenever $Q(n)$ is true.

By the principle of mathematical induction, $Q(n)$ is true for all $n \in \mathbb{N}$.

□

Lemma 2.3.7. *Let J_m, J_n be given for some $m, n \in \mathbb{N}$. Then we have the following properties:*

(A) *If $m \leq n$, then there exists an injection $f : J_m \to J_n$.*

(B) *If $m \geq n$, then there exists a surjection $f : J_m \to J_n$.*

(C) *If $m = n$, then there exists a bijection $f : J_m \to J_n$.*

This lemma is the converse of Lemma 2.3.6.

Proof. We will prove this lemma by explicitly constructing the functions.

(A) If $m \leq n$, let $f : J_m \to J_n$ be a function defined by

$$f(i) = i \quad \text{for all } i \in J_m.$$

Then f is an injection.

(B) If $m \geq n$, let $f : J_m \to J_n$ be a function defined by

$$f(i) = \begin{cases} i, & \text{if } i \leq n \\ n, & \text{if } i > n \end{cases}$$

Then f is a surjection.

(C) If $m = n$, let $f : J_m \to J_n$ be a function defined by

$$f(i) = i \quad \text{for all } i \in J_m.$$

Then f is a bijection.

$\square$

Theorem 2.3.8. *Let X, Y be finite sets. Then we have the following properties:*

*(**A**) If there exists an injection $f : X \to Y$, then we have $|X| \leq |Y|$.*

*(**B**) If there exists a surjection $f : X \to Y$, then we have $|X| \geq |Y|$.*

*(**C**) If there exists a bijection $f : X \to Y$, then we have $|X| = |Y|$.*

*Moreover, the converses of (**A**), (**B**), and (**C**) are also true.*

*The contraposition of (**A**) is called the **pigeonhole principle**, which states that if $|X| > |Y|$, then there exists no injection $f : X \to Y$.*

Proof. Let $|X| = m$ and $|Y| = n$ for some $m, n \in \mathbb{N}$. Then there exist bijections $f_X : J_m \to X$ and $f_Y : J_n \to Y$. By Theorems 2.2.3 and 2.2.7, the composition

$$F \equiv (f_Y)^{-1} \circ f \circ f_X : J_m \to J_n$$

is injective, surjective, or bijective if f is injective, surjective, or bijective, respectively. Then, by Lemma 2.3.6, we have the desired results:

(A) If f is injective, then F is injective, which implies that $m \leq n$.

(B) If f is surjective, then F is surjective, which implies that $m \geq n$.

(C) If f is bijective, then F is bijective, which implies that $m = n$.

Also, by Theorems 2.2.3 and 2.2.7, the composition

$$f \equiv f_Y \circ G \circ (f_X)^{-1} : X \to Y$$

is injective, surjective, or bijective if $G : J_m \to J_n$ is injective, surjective, or bijective, respectively. Then, by Lemma 2.3.7, we have the desired converses:

(A) If $m \leq n$, then there exists an injective function $G : J_m \to J_n$, which implies that there exists an injective function $f : X \to Y$.

(B) If $m \geq n$, then there exists a surjective function $G : J_m \to J_n$, which implies that there exists a surjective function $f : X \to Y$.

(C) If $m = n$, then there exists a bijective function $G : J_m \to J_n$, which implies that there exists a bijective function $f : X \to Y$.

□

Lemma 2.3.9. *Let X, Y be finite sets with $X \cap Y = \emptyset$. Then the union $X \cup Y$ is finite.*

Proof. Let $m = |X|$ and $n = |Y|$. There exist bijections $f : J_m \to X$ and $g : J_n \to Y$. Let $h : J_{m+n} \to X \cup Y$ be a function defined by

$$h(i) = \begin{cases} f(i), & \text{if } i \leq m \\ g(i-m), & \text{if } m < i \leq m+n \end{cases}$$

Then h is bijective, which implies that $X \cup Y$ is finite. □

Theorem 2.3.10. *Let X, Y be finite sets. Then the union $X \cup Y$ and intersection $X \cap Y$ are finite.*

Proof. Let A, B, C be sets defined by

- $A = X \setminus Y \subseteq X$.
- $B = Y \setminus X \subseteq Y$.
- $C = X \cap Y \subseteq X$.

Then A, B, C are finite by Theorem 2.3.2 and are disjoint each other. We have the followings:

- $X = (X \setminus Y) \cup (X \cap Y) = A \cup C$.
- $Y = (Y \setminus X) \cup (X \cap Y) = B \cup C$.
- $X \cup Y = ((X \setminus Y) \cup (Y \setminus X)) \cup (X \cap Y) = (A \cup B) \cup C$.

By Lemma 2.3.9, $X, Y, X \cup Y$ are finite. □

Theorem 2.3.11. *Let $X_1, X_2, \ldots, X_n$ be finite sets for some $n \in \mathbb{N}$. Then the union $\bigcup_{k=1}^{n} X_k$ and intersection $\bigcap_{k=1}^{n} X_k$ are finite.*

Proof. $\bigcap_{k=1}^{n} X_k \subseteq X_1$ is finite by Theorem 2.3.2 for any $n \in \mathbb{N}$. Let $P(n)$ be the statement that

$$X_1, X_2, \dots, X_n \text{ are finite sets } \implies \bigcup_{k=1}^{n} X_k \text{ is finite.}$$

- **Base case:** Let X_1 be a finite set. Then $\bigcup_{k=1}^{1} X_k = X_1$ is finite. Thus $P(1)$ is true.
- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$.
  Let $X_1, X_2, \dots, X_{n+1}$ be finite sets. Then we have
  $$\bigcup_{k=1}^{n+1} X_k = \left(\bigcup_{k=1}^{n} X_k \right) \cup X_{n+1}.$$
  By the inductive hypothesis, $\bigcup_{k=1}^{n} X_k$ is finite. Then, by Theorem 2.3.10, $\bigcup_{k=1}^{n+1} X_k$ is finite. Thus $P(n+1)$ is true whenever $P(n)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$. □

Lemma 2.3.12. *Let X, Y be finite sets. Then the cartesian product $X \times Y$ is finite.*

Proof. Let $m = |X|$ and $n = |Y|$. There exist bijections $f : J_m \to X$ and $g : J_n \to Y$. Let $h : J_{mn} \to X \times Y$ be a function defined by

$$h(i) = \left(f\left(\left\lceil \frac{i}{n} \right\rceil \right), g\left(i - n\left(\left\lceil \frac{i}{n} \right\rceil - 1 \right) \right) \right) \text{ for all } i \in J_{mn}.$$

Then h is bijective, which implies that $X \times Y$ is finite. □

Theorem 2.3.13. *Let $X_1, X_2, \dots, X_n$ be finite sets for some $n \in \mathbb{N}$. Then the cartesian product $\prod_{k=1}^{n} X_k$ is finite.*

Proof. Let $P(n)$ be the statement that

$$X_1, X_2, \dots, X_n \text{ are finite sets } \implies \prod_{k=1}^{n} X_k \text{ is finite.}$$

- **Base case:** Let X_1 be a finite set. Then, $\prod_{k=1}^{1} X_k = X_1$ is finite. Thus $P(1)$ is true.
- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$.

  Let $X_1, X_2, \ldots, X_{n+1}$ be finite sets. Then we have
  $$\prod_{k=1}^{n+1} X_k = \left(\prod_{k=1}^{n} X_k\right) \times X_{n+1}.$$
  By the inductive hypothesis, $\prod_{k=1}^{n} X_k$ is finite. Then, by Lemma 2.3.12, $\prod_{k=1}^{n+1} X_k$ is finite. Thus $P(n+1)$ is true whenever $P(n)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$. $\square$

Theorem 2.3.14. *Let X be a nonempty set. Then there exists no surjection $f : X \to \mathcal{P}(X)$, where*
$$\mathcal{P}(X) := \{E \mid E \subseteq X\}$$
*is called the **power set** of X. This theorem is called **Cantor's theorem**.*

Proof. Let S be a set defined by
$$S := \{x \in X \mid x \notin f(x)\}.$$
Since $S \subseteq X$, we have that $S \in \mathcal{P}(X)$. Assume that there exists a surjection $f : X \to \mathcal{P}(X)$. Then there exists $x \in X$ such that $f(x) = S$.

If $x \in S$, then we have $x \notin f(x) = S$, which is a contradiction. If $x \notin S$, then we have $x \in f(x) = S$, which is also a contradiction. Therefore, there exists no surjection $f : X \to \mathcal{P}(X)$. $\square$

2.4 Countable and Uncountable Sets

A set X is said to be **countable** if $\mathbb{N} \sim X$. A set X is said to be **uncountable** if it is neither finite nor countable. A set X is said to be **at most countable** if X is either finite or countable.

Theorem 2.4.1. *Let X be an infinite set, and let Y be a finite set. Then the set $X \setminus Y$ is infinite.*

Proof. Assume that $X \setminus Y$ is finite. Since Y is finite, by Theorem 2.3.10, $(X \setminus Y) \cup Y$ is finite. Then, by Theorem 2.3.2, $X \subseteq (X \setminus Y) \cup Y$ is finite, which contradicts the assumption. Therefore, $X \setminus Y$ is infinite. □

Theorem 2.4.2. *Let X be an infinite set. If there exists a injection $f : X \to Y$, then Y is infinite.*

Proof. Assume that Y is finite. Let $g : Y \to X$ be defined by

$$g(y) = \begin{cases} f^{-1}(y), & \text{if } y \in f(X) \\ f^{-1}(y_0), & \text{Otherwise} \end{cases}$$

where $y_0 \in f(X)$ is fixed. Then g is surjective. By Theorem 2.3.4, X is finite, which is a contradiction. Therefore, Y is infinite. □

Lemma 2.4.3. $\mathbb{N}$ *is infinite.*

Proof. Assume that $\mathbb{N}$ is finite. Then there exists a bijection $f : J_n \to \mathbb{N}$ for some $n \in \mathbb{N}$. Then, by Theorem 2.3.2, $f(J_n) \subseteq \mathbb{N}$ is finite. By Theorem 2.3.5, $f(J_n)$ has a greatest element, say m.

Then we have $m+1 \in \mathbb{N}$ and $m < m+1 \notin f(J_n)$, which contradicts the assumption that f is surjective. Therefore, $\mathbb{N}$ is infinite. □

Theorem 2.4.4. *Any countable set is infinite.*

Proof. Let X be a countable set. Then there exists a bijection $f : \mathbb{N} \to X$. By Theorem 2.4.2 and lemma 2.4.3, X is infinite. □

Remark. Let X be a countable set. Then there exists a bijection $f : \mathbb{N} \to X$. The elements of X can be listed as $X = \{x_1, x_2, x_3, \ldots\}$, where $x_n = f(n)$ for all $n \in \mathbb{N}$. In other words, the elements of X can be arranged in a sequence.

Theorem 2.4.5. $\mathbb{Z}$ *is countable.*

Proof. Let $f : \mathbb{N} \to \mathbb{Z}$ be a function defined by

$$f(n) = \begin{cases} -\dfrac{n}{2}, & \text{if } n \text{ is even} \\ \dfrac{n-1}{2}, & \text{if } n \text{ is odd} \end{cases}$$

We will show that f is bijective, which implies that $\mathbb{Z}$ is countable.

- **Injectivity:** Assume that $f(n_1) = f(n_2)$ for some $n_1, n_2 \in \mathbb{N}$. Since $f(n) \geq 0$ for odd n and $f(n) < 0$ for even n, we have n_1 and n_2 are either both even or both odd.
  - If both n_1 and n_2 are even, then we have $-\frac{n_1}{2} = -\frac{n_2}{2}$, which implies $n_1 = n_2$.
  - If both n_1 and n_2 are odd, then we have $\frac{n_1 - 1}{2} = \frac{n_2 - 1}{2}$, which implies $n_1 = n_2$.

  Thus f is injective.
- **Surjectivity:** Let $k \in \mathbb{Z}$ be given. We need to find an $n \in \mathbb{N}$ such that $f(n) = k$.
  - If $k \geq 0$, let $n = 2k + 1$. Then we have $f(n) = \frac{n-1}{2} = \frac{2k}{2} = k$.
  - If $k < 0$, let $n = -2k$. Then we have $f(n) = -\frac{n}{2} = -\frac{-2k}{2} = k$.

  Thus f is surjective.

□

Lemma 2.4.6. *Any infinite subset of* $\mathbb{N}$ *is countable.*

Proof. Let $X \subseteq \mathbb{N}$ be an infinite subset. Let $\{X_n\}_{n=1}^{\infty}$ be a sequence and let $f : \mathbb{N} \to X$ be a function defined as follows:

- $X_1 \coloneqq X$, $f(1) = \min X_1$
- $X_2 \coloneqq X_1 \setminus \{f(1)\}$, $f(2) = \min X_2$

  $\vdots$
- $X_{n+1} \coloneqq X_n \setminus \{f(n)\}$, $f(n+1) = \min X_{n+1}$

for each $n \in \mathbb{N}$. By Theorem 2.4.1, X_n is infinite for each $n \in \mathbb{N}$, so is nonempty. Then, by Theorem 1.1.1, X_n has a least element, so f is well-defined. We will show that f is bijective, which implies that X is countable.

- **Injectivity:** Assume that $n_1 \neq n_2$ for some $n_1, n_2 \in \mathbb{N}$. Without loss of generality, assume that $n_1 < n_2$. Then we have

$$f(n_1) \in X_{n_1} \text{ and } f(n_2) \in X_{n_2}.$$

  Since $X_{n_2} \subseteq X_{n_1} \setminus \{f(n_1)\}$, we have $f(n_2) \neq f(n_1)$. Thus f is injective.

- **Surjectivity:** Let $x \in X$ be given. Let S be a set defined as

$$S := \{n \in X \mid n < x\}.$$

If $S = \emptyset$, then $f(1) = x$. Otherwise, by Lemma 2.3.1, $S \subseteq J_{x-1}$ is finite. Let $k = |S|$, $S = \{s_1, s_2, \dots, s_k\}$ with $s_1 < s_2 < \cdots < s_k$. Then we have

  - $\min X_1 = \min S \implies X_2 = X \setminus \{s_1\}$.
  - $\min X_2 = \min S \setminus \{s_1\} \implies X_3 = X \setminus \{s_1, s_2\}$.

    $\vdots$
  - $\min X_k = \min S \setminus \{s_1, s_2, \dots, s_{k-1}\} \implies X_{k+1} = X \setminus S$.

Thus $f(k+1) = \min X_{k+1} = x$. Therefore, f is surjective.

$\square$

Theorem 2.4.7. *Any infinite subset of a countable set is countable.*

Proof. Let X be a countable set and let $Y \subseteq X$ be an infinite subset. Since X is countable, there exists a bijection $f : \mathbb{N} \to X$. By Theorem 2.2.5, $f|_{f^{-1}(Y)}^{Y} : f^{-1}(Y) \to Y$ is bijective. Then, by Theorem 2.3.4, $f^{-1}(Y) \subseteq \mathbb{N}$ is infinite.

By Lemma 2.4.6, $f^{-1}(Y)$ is countable. Then there exists a bijection $g : \mathbb{N} \to f^{-1}(Y)$. The composition $f|_{f^{-1}(Y)}^{Y} \circ g : \mathbb{N} \to Y$ is bijective, which implies that Y is countable. $\square$

Theorem 2.4.8. *Let X be infinite. If there exists a surjection $f : \mathbb{N} \to X$, then X is countable.*

Proof. For each $k \in X$, let S_k be a set defined as

$$S_k := f^{-1}(\{k\}).$$

Since f is surjective, $S_k \neq \emptyset$ and by Theorem 1.1.1, S_k has a least element for all $k \in X$. Let $g : X \to \mathbb{N}$ be a function defined by

$$g(k) = \min S_k \text{ for all } k \in X.$$

Assume that $g(k_1) = g(k_2)$ for some $k_1, k_2 \in X$, i.e., $\min S_{k_1} = \min S_{k_2}$. Since $S_i \cap S_j = \emptyset$ for all $i, j \in X$ with $i \neq j$, we have $S_{k_1} = S_{k_2} \implies k_1 = k_2$. Thus g is injective.

Since g is injective, by Theorem 2.2.5, $g|^{g(X)} : X \to g(X)$ is bijective. Since X is infinite, $g(X) \subseteq \mathbb{N}$ is infinite by Theorem 2.4.2. Then, by Lemma 2.4.6, $g(X)$ is countable, which implies that X is countable. $\square$

Theorem 2.4.9. *Let $\{X_n\}_{n\in\mathbb{N}}$ be a sequence of countable sets. Then $\bigcup_{n=1}^{\infty} X_n$ is countable, i.e., any countable union of countable sets is countable.*

Proof. There exists a bijection $f_n : \mathbb{N} \to X_n$ for each $n \in \mathbb{N}$. For each $n \in \mathbb{N}$, we define the followings:

- $t_n := \min T_n,$ where $T_n := \left\{ k \in \mathbb{N} \,\middle|\, n \leq \dfrac{k(k+1)}{2} \right\}.$
- $s_n := n - \dfrac{t_n(t_n-1)}{2} = n - \dfrac{t_n(t_n+1)}{2} + t_n \in \mathbb{N}.$

Since $2n \in T_n$, T_n is nonempty. Then, by Theorem 1.1.1, T_n has a least element, so t_n is well-defined. Since $\dfrac{t_n(t_n-1)}{2} < n \leq \dfrac{t_n(t_n+1)}{2}$, we have

$$t_n - s_n + 1 = \frac{t_n(t_n+1)}{2} - n + 1 \geq 1,$$

so $t_n - s_n + 1 \in \mathbb{N}$. Let $g : \mathbb{N} \to \bigcup_{n=1}^{\infty} X_n$ be a function defined by

$$g(k) := f_{s_k}(t_k - s_k + 1) \text{ for all } k \in \mathbb{N}.$$

Let $x \in \bigcup_{n=1}^{\infty} X_n$ be given. Then there exists some $m \in \mathbb{N}$ such that $x \in X_m$. Since f_m is surjective, there exists some $l \in \mathbb{N}$ such that $f_m(l) = x$. Let

$$k := \frac{(l+m-1)(l+m-2)}{2} + m \in \mathbb{N}.$$

We will show that $l + m - 2 \notin T_k$ and $l + m - 1 \in T_k$, which implies that

- $t_k = \min T_k = l + m - 1.$
- $s_k = k - \dfrac{t_k(t_k-1)}{2} = m.$
- $g(k) = f_{s_k}(t_k - s_k + 1) = f_m(l + m - 1 - m + 1) = f_m(l) = x.$

Consider the following inequalities when $z = l + m - 2$ and $z = l + m - 1$:

- Let $z = l + m - 2$. Then we have

$$\begin{aligned} k - \frac{z(z+1)}{2} &= k - \frac{(l+m-2)(l+m-1)}{2} \\ &= \frac{(l+m-1)(l+m-2)}{2} + m - \frac{(l+m-2)(l+m-1)}{2} \\ &= m > 0. \end{aligned}$$

- Let $z = l + m - 1$. Then we have

$$\begin{aligned} k - \frac{z(z+1)}{2} &= k - \frac{(l+m-1)(l+m)}{2} \\ &= \frac{(l+m-1)(l+m-2)}{2} + m - \frac{(l+m-1)(l+m)}{2} \\ &= m - \frac{(l+m-1)(l+m) - (l+m-1)(l+m-2)}{2} \\ &= m - (l+m-1) = 1 - l \le 0. \end{aligned}$$

Thus g is surjective, which implies that $\bigcup_{n=1}^{\infty} X_n$ is countable by Theorem 2.4.8. □

Corollary 2.4.10. *Let $X_1, X_2, \ldots, X_n$ be countable sets for some $n \in \mathbb{N}$. Then $\bigcup_{i=1}^{n} X_i$ is countable, i.e., any finite union of countable sets is countable.*

Proof. Let $X_{n+1}, X_{n+2}, \ldots$ be countable sets. Then, by Theorem 2.4.9, $\bigcup_{n=1}^{\infty} X_n$ is countable. Assume that $\bigcup_{i=1}^{n} X_i$ is finite. Then $X_1 \subseteq \bigcup_{i=1}^{n} X_i$ is finite by Theorem 2.3.2, which contradicts Theorem 2.4.4. Thus $\bigcup_{i=1}^{n} X_i$ is infinite. Since $\bigcup_{i=1}^{n} X_i \subseteq \bigcup_{n=1}^{\infty} X_n$, by Theorem 2.4.7, $\bigcup_{i=1}^{n} X_i$ is countable. □

Theorem 2.4.11. *Let X be a countable set. If Y_x is finite for each $x \in X$, then the union*

$$Y = \bigcup_{x \in X} Y_x$$

is countable, i.e., any countable union of finite sets is countable.

Proof. There exists a bijection $f : \mathbb{N} \to X$. Also, for each $x \in X$, there exists a bijection $g_x : J_{n_x} \to Y_x$ for some $n_x \in \mathbb{N}$. Let $h : \mathbb{N} \to Y$ be a function defined by

- $h(j) := g_{f(1)}(j)$ for $j = 1, 2, \cdots, n_{f(1)}$.
- $h\left(\sum_{i=1}^{k-1} n_{f(i)} + m\right) := g_{f(k)}(m)$ for each $k \in \mathbb{N}$ with $k \ge 2$, and $m = 1, 2, \cdots, n_{f(k)}$.

Then h is surjective. Therefore, by Theorem 2.4.8, Y is countable. $\square$

Lemma 2.4.12. $\mathbb{N} \times \mathbb{N}$ *is countable.*

Proof. We will prove this lemma by explicitly constructing a bijection from $\mathbb{N}\times\mathbb{N} \to \mathbb{N}$. Let $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ be a function defined by

$$f(m,n) := \frac{(m+n-1)(m+n-2)}{2} + m$$

for each $(m,n) \in \mathbb{N} \times \mathbb{N}$. We will show that f is bijective.

- **Injectivity:** Assume that $(m_1,n_1) \neq (m_2,n_2)$. If $m_1+n_1 = m_2+n_2$, then $m_1 \neq m_2$, so $f(m_1,n_1) \neq f(m_2,n_2)$. Otherwise, $m_1+n_1 \neq m_2+n_2$.

  Without loss of generality, assume $m_1+n_1 < m_2+n_2$. Then, since $m_1 + n_1 + 1 \leq m_2+n_2$, we have

$$\begin{aligned} f(m_1,n_1) &= \frac{(m_1+n_1-1)(m_1+n_1-2)}{2} + m_1 \\ &\leq \frac{(m_1+n_1-1)(m_1+n_1-2)}{2} + (m_1+n_1-1) \\ &= \frac{(m_1+n_1)(m_1+n_1-1)}{2} \\ &\leq \frac{(m_2+n_2-1)(m_2+n_2-2)}{2} \\ &< \frac{(m_2+n_2-1)(m_2+n_2-2)}{2} + m_2 = f(m_2,n_2), \end{aligned}$$

  Thus $f(m_1,n_1) \neq f(m_2,n_2)$, so f is injective.

- **Surjectivity:** Let $k \in \mathbb{N}$ be given. Let t_k, s_k be defined as in the proof of Theorem 2.4.9. Then we have $f(s_k, t_k - s_k + 1) = k$. Thus f is surjective.

$\square$

Theorem 2.4.13. *Let X and Y be countable sets. Then the cartesian product $X \times Y$ is countable.*

Proof. There exist bijections $f : \mathbb{N} \to X$ and $g : \mathbb{N} \to Y$. Let $h : \mathbb{N}\times\mathbb{N} \to X \times Y$ be a function defined by

$$h(m,n) := (f(m), g(n)) \text{ for all } (m,n) \in \mathbb{N} \times \mathbb{N}.$$

Then h is bijective. By Lemma 2.4.12, $X \times Y$ is countable. $\square$

Theorem 2.4.14. *Let $X_1, X_2, \ldots, X_n$ be countable sets for some $n \in \mathbb{N}$. Then $\prod_{i=1}^{n} X_i$ is countable, i.e., any finite product of countable sets is countable.*

Proof. Let $P(n)$ be the statement that

$$X_1, X_2, \ldots, X_n \text{ are countable sets } \implies \prod_{i=1}^{n} X_i \text{ is countable.}$$

- **Base case:** Let X_1 be a countable set. Then $\prod_{i=1}^{1} X_i = X_1$ is countable. Thus $P(1)$ is true.
- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$.

  Let $X_1, X_2, \ldots, X_{n+1}$ be countable sets. Then we have

  $$\prod_{i=1}^{n+1} X_i = \left(\prod_{i=1}^{n} X_i \right) \times X_{n+1}$$

  By the inductive hypothesis, $\prod_{i=1}^{n} X_i$ is countable. Then, by Theorem 2.4.13, $\prod_{i=1}^{n+1} X_i$ is countable. Thus $P(n+1)$ is true when $P(n)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$. □

Theorem 2.4.15. *$\mathbb{Q}$ is countable.*

Proof. Any $q \in \mathbb{Q}$ can be expressed as $q = \frac{m}{n}$ for some $m \in \mathbb{Z}$ and $n \in \mathbb{N}$. Let $f : \mathbb{Z} \times \mathbb{N} \to \mathbb{Q}$ be a function defined by

$$f(m, n) := \frac{m}{n} \text{ for all } (m, n) \in \mathbb{Z} \times \mathbb{N}.$$

Then f is surjective. By Theorems 2.4.5 and 2.4.13, $\mathbb{Z} \times \mathbb{N}$ is countable. Then there exists a bijection $g : \mathbb{N} \to \mathbb{Z} \times \mathbb{N}$.

By Theorem 2.2.7, $f \circ g : \mathbb{N} \to \mathbb{Q}$ is surjective. Since $\mathbb{N} \subseteq \mathbb{Q}$, by Lemma 2.4.3 and theorem 2.3.2, $\mathbb{Q}$ is infinite. Therefore, by Theorem 2.4.8, $\mathbb{Q}$ is countable. □

Lemma 2.4.16. $X := \prod_{n=1}^{\infty} \{0, 1\}$ *is uncountable.*

Proof. For each $n \in \mathbb{N}$, let $e_n \in X$ be a sequence defined by

$$e_{nm} := \begin{cases} 1 & \text{if } m = n \\ 0 & \text{Otherwise} \end{cases}$$

Let $f : \mathbb{N} \to X$ be a function defined by

$$f(n) = e_n \text{ for all } n \in \mathbb{N}.$$

Then f is injective. By Lemma 2.4.3 and theorem 2.4.2, X is infinite.

Assume that X is countable. Then, there exists a bijection $g : \mathbb{N} \to X$. Let $X = \{g(1), g(2), g(3), \ldots\}$, where

- $g(1) = (a_{11}, a_{12}, a_{13}, \ldots)$.
- $g(2) = (a_{21}, a_{22}, a_{23}, \ldots)$.

  $\vdots$

Let $x = (b_1, b_2, b_3, \ldots) \in X$ be a sequence defined by

$$b_n := 1 - a_{nn} \text{ for all } n \in \mathbb{N}.$$

Then, $x \neq g(n)$ for all $n \in \mathbb{N}$, which contradicts the assumption that g is surjective. Therefore, X is uncountable. □

Theorem 2.4.17. $\mathbb{R}$ *is uncountable.*

Proof. Since $\mathbb{N} \subseteq \mathbb{R}$, by Lemma 2.4.3 and theorem 2.3.2, $\mathbb{R}$ is infinite. Let $f : \mathbb{N} \to [0, 1)$ be a function defined by

$$f(n) := \frac{1}{n+1} \text{ for all } n \in \mathbb{N}.$$

Then f is injective. By Lemma 2.4.3 and theorem 2.4.2, $[0, 1)$ is infinite. We will prove that $[0, 1) \subseteq \mathbb{R}$ is uncountable, which implies that $\mathbb{R}$ is uncountable by Theorem 2.4.7.

Let $g : [0, 1) \to \prod_{n=1}^{\infty} \{0, 1\}$ be a function defined by

$$g(x) := \{a_n^x\}_{n=1}^{\infty} \text{ for all } x \in [0, 1)$$

where $\{a_n^x\}_{n=1}^{\infty}$ is the *greedy base* 2 *expansion* of x. Let Y be a set defined by

$$Y := g([0, 1)) \subseteq \prod_{n=1}^{\infty} \{0, 1\}.$$

Then, by Theorems 1.3.9 and 2.2.5, $g|^Y$ is bijective.

Assume that $[0,1)$ is countable. Then there exists a bijection $h : \mathbb{N} \to [0,1)$. By Theorem 2.2.7, $g|^Y \circ h : \mathbb{N} \to Y$ is bijective, so Y is countable. Let $Z :=$ $\prod_{n=1}^{\infty}\{0,1\} \setminus Y$. By Theorems 1.3.10 and 1.3.11, we have

$$Z = \left\{ \{a_n\}_{n=1}^{\infty} \in \prod_{n=1}^{\infty}\{0,1\} \;\middle|\; a_n = 1 \text{ for all } n \geq N \text{ for some } N \in \mathbb{N} \right\}.$$

Let Z_N be a set defined by

$$Z_N := \left\{ \{a_n\}_{n=1}^{\infty} \in \prod_{n=1}^{\infty}\{0,1\} \;\middle|\; a_n = 1 \text{ for all } n \geq N \right\} \text{ for each } N \in \mathbb{N}.$$

Let $F_N : \prod_{n=1}^{N-1}\{0,1\} \to Z_N$ be a function defined by

$$F_N\left(\{a_n\}_{n=1}^{N-1}\right) := \{a_n\}_{n=1}^{N-1} \cup \{1\}_{n=N}^{\infty} \text{ for all } \{a_n\}_{n=1}^{N-1} \in \prod_{n=1}^{N-1}\{0,1\}.$$

Then F_N is surjective. By Theorems 2.3.4 and 2.3.13, Z_N is finite. Then, by Theorem 2.4.11, $Z = \bigcup_{N=1}^{\infty} Z_N$ is countable.

By Corollary 2.4.10, $\prod_{n=1}^{\infty}\{0,1\} = Y \cup Z$ is countable. Lemma 2.4.16 shows that this is a contradiction. Therefore, $[0,1)$ is uncountable. $\square$

2.5 Metric Spaces

Let X be a set. A function $d : X \times X \to \mathbb{R}$ is called a **metric** on X if for any $p, q \in X$, we have

(A) $d(p,q) \geq 0$ and $d(p,q) = 0 \iff p = q$.

(B) $d(p,q) = d(q,p)$.

(C) $d(p,r) \leq d(p,q) + d(q,r)$ for any $r \in X$.

(C) is called the **triangle inequality**. The pair (X,d) is called a **metric space**.

Theorem 2.5.1. *Let $d : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ be a function defined by*

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| = \sqrt{\sum_{k=1}^{n} (x_k - y_k)^2}$$

*for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Then d is a metric on $\mathbb{R}^n$. Especially, d is called the **standard metric** on $\mathbb{R}^n$.*

Proof. Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ be given.

- $d(\mathbf{x}, \mathbf{y}) \geq 0$ is obvious. Also, we have

$$\begin{aligned} d(\mathbf{x}, \mathbf{y}) = 0 &\iff \|\mathbf{x} - \mathbf{y}\| = 0 \\ &\iff x_k = y_k \text{ for all } k = 1, 2, \ldots, n \\ &\iff \mathbf{x} = \mathbf{y}. \end{aligned}$$

- $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$ is obvious.

- By Theorem 1.6.2, we have

$$\begin{aligned} d(\mathbf{x}, \mathbf{z}) &= \|\mathbf{x} - \mathbf{z}\| = \|(\mathbf{x} - \mathbf{y}) + (\mathbf{y} - \mathbf{z})\| \\ &\leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y} - \mathbf{z}\| = d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z}) \end{aligned}$$

Therefore, d is a metric on $\mathbb{R}^n$. ☐

Definition 2.5.2. Let (X, d) be a metric space. For any $p \in X$ and $r > 0$, we define the **neighborhood** of p with radius r as

$$N_r(p) := \{q \in X \mid d(p, q) < r\}.$$

Especially, if $X = \mathbb{R}^n$ and d is the standard metric on $\mathbb{R}^n$, then $N_r(p)$ is called the **open ball** of radius r centered at p, which is denoted by $B_r(p)$.

Definition 2.5.3. Let (X, d) be a metric space and $E \subseteq X$. Let $p \in X$ be given.

- p is called an **limit point** of E if for any $r > 0$, we have $E \cap (N_r(p) \setminus \{p\}) \neq \emptyset$.
- p is called a **isolated point** of E if there exists some $r > 0$ such that $N_r(p) \cap E = \{p\}$.
- p is called an **interior point** of E if there exists some $r > 0$ such that $N_r(p) \subseteq E$.

Definition 2.5.4. Let (X, d) be a metric space and $E \subseteq X$.

- E is called **open** if Any point of E is an interior point of E.
- E is called **closed** if Any limit point of E belongs to E.

Theorem 2.5.5. *Let (X, d) be a metric space. For any $p \in X$ and $r > 0$, the neighborhood $N_r(p)$ is open.*

Proof. Let $q \in N_r(p)$ be given. Then we have $d(p, q) < r$. Put $\varepsilon = r - d(p, q) > 0$. Let $s \in N_\varepsilon(q)$ be given. Then we have $d(q, s) < \varepsilon$ and

$$d(p, s) \leq d(p, q) + d(q, s) < d(p, q) + \varepsilon = r.$$

Thus $s \in N_r(p)$, which implies that $N_\varepsilon(q) \subseteq N_r(p)$. Since q is arbtrary, $N_r(p)$ is open. $\square$

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